



Currents in the high drag environment of a coastal kelp stand off California

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Abstract—The physical environment within a California kelp bed is modified from that of the region by the large drag of the massed plants. The change is greatest for longshore motions, where velocities diminish with distance upcoast of the leading edge of the bed in a manner consistent with the damping expected for coastally-trapped waves. Cross-shore velocity fluctuations are more important for the movement of material between offshore waters and the kelp bed interior. Results suggest that penetration of water for distances of 400 m into the kelp is common. The importance of water motions for ecological processes ranging from enhancing boundary layer transport to kelp blades to moving planktonic larvae implies that size of a kelp bed and position within it are important for the kelp ecosystem. © 1998 Elsevier Science Ltd. All rights reserved

1. INTRODUCTION

The kelp beds off the coast of Southern California are important ecological systems that are large enough to affect their physical environments. Because kelp beds depend on the movement of plants, animals, and nutrients to and from the surrounding oceanic waters, understanding the nature of the currents within and the exchange with waters outside enables us to understand kelp bed ecological interactions.

Physically, kelp beds are regions of enhanced hydrodynamic drag (Jackson and Winant, 1983). The bed off Point Loma, San Diego (Fig. 1), is one of the largest along the California coast, having a width of 1–1.5 km that is comparable to an internal Rossby radius (~ 1.5 km in this shallow water) and a length of 7 km that is comparable to wavelengths of suprainertial frequency Kelvin waves (Jackson, 1984, 1988). Because of its large relative size, a kelp bed has the potential for affecting the propagation of coastally-trapped waves by slowing and damping them as they propagate along the coast (Jackson, 1988). As a result, currents should have different properties in the region of a kelp bed than in a similar kelp-free region.

A kelp plant grows attached to hard bottom by a root-like holdfast. Multiple vine-like fronds grow upward from each holdfast, buoyed by small air bladders (pneumatocysts). At the surface, fronds spread out to form a canopy which can totally cover the surface. Each frond consists of a rope-like stipe, about 1 cm in diameter, to which are connected floats containing blades. The fronds of an individual plant bunch together to form a compact column between the bottom and the surface that typically may contain 40 fronds and is between 10 and 20 cm in diameter. Plant densities at the Pt. Loma bed range from 0.01 to 0.1 plants m^{-2} , with lower densities typically in deeper water. The drag in this kelp stand is a

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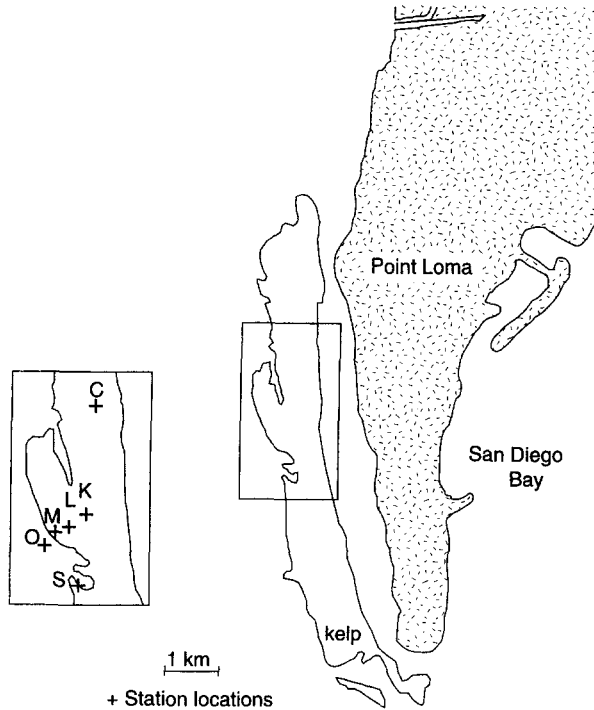


Fig. 1. Map of current meter mooring locations off Point Loma, San Diego, California.

factor of 10 greater than that in a non-kelp area and the currents are slower (Jackson and Winant, 1983). Kelp plant drag affects internal waves propagating across the bed, damping the high frequency waves and slowing the low frequency waves (Jackson, 1984). Kelp beds also dampen surface gravity waves propagating through them (Elwany *et al.*, 1994), at a cost to the kelp that includes the ripping out of plants by the forces resulting from the high drag (Seymour *et al.*, 1989).

In this paper, the patterns of velocities and temperatures in the Point Loma kelp bed and their significance for kelp ecology are discussed.

2. OBSERVATIONS

The Pt. Loma kelp bed is situated over a gently sloping rock shelf which reaches 22 m bottom depth 1600 m from shore, drops to 30 m in an additional 100 m offshore, and then merges into a shallow, downward-sloping sandy bottom. The kelp bed grows to the surface from depths of ~ 6 –30 m.

Temperature and current measurements were made using vector measuring current meters (VMCMs) on taut line moorings. The instruments recorded average north and eastward currents and temperature every 4 min. Further discussion of the instruments is given by Winant and Bratkovich (1981).

Records exist for five deployments (Table 1 and Fig. 1), but the results emphasized in this paper are for periods with more extensive longshore or cross-shore coverage: 14 September

Table 1. Summary statistics for velocity measurements

Parameter	Location							
	KT	KM	KB	L	M	O	S	C
Instrument depth (m)	5	7	11	5	8	6	5	5
Bottom depth (m)	14	14	14	14	17	28	14	14
Distance N (km)	4.34	4.34	4.34	4.24	4.18	4.06	3.44	5.69
Distance E (km)	-1.18	-1.18	-1.18	-1.38	-1.58	-1.75	-1.48	-1.04
Distance from edge (km)	0.53	0.53	0.53	0.31	0.12	-0.07	0.04	0.43
Deploy. 1: 2-10 Dec 1980								
u (cm s ⁻¹)	Mean		0.0			0.2		
	SD		0.5			2.2		
v (cm s ⁻¹)	Mean		0.0			-0.5		
	SD		1.7			5.9		
Deploy. 2: 13-21 Feb 1981								
u (cm s ⁻¹)	Mean	0.6	0.8	0.7				
	SD	1.4	1.4	1.5				
v (cm s ⁻¹)	Mean	-2.6	-3.0	-2.1				
	SD	3.5	4.0	3.5				
Deploy. 3: 2-14 Sept 1981								
u (cm s ⁻¹)	Mean	-0.8	0.2					
	SD	1.8	0.7					
v (cm s ⁻¹)	Mean	-0.1	-0.8					
	SD	1.5	1.5					
Deploy. 4: 14 Sept-4 Oct 1981								
u (cm s ⁻¹)	Mean		-0.4	0.5		-0.1	0.9	0.3
	SD		1.3	1.5		4.3	1.9	1.1
v (cm s ⁻¹)	Mean		0.2	-0.4		2.0	-0.1	-0.3
	SD		1.3	2.2		8.8	3.0	1.1
Deploy. 6,7: 24 Sept-21 Oct 1982								
u (cm s ⁻¹)	Mean	0.3		-0.2	0.4	0.3	0.9	
	SD	0.9		1.1	1.6	3.6	2.2	
v (cm s ⁻¹)	Mean	-0.3		0.2	1.4	2.1	-0.2	
	SD	2.0		2.0	5.3	9.9	3.9	

to 4 October 1981 (Deployment 4) and 24 September to 21 October 1982 (Deployment 7). There were four moorings located on a straight line perpendicular to shore over a total distance of 600 m, with instruments separated by intervals of about 200 m. Additional moorings were located about 0.9 km south of this line 40 m shoreward of the outer edge and 1.3 km north of this line, about 430 m shoreward of the outer edge (Fig. 1). There were more instruments deployed during Deployment 4 but a large number of instrument failures made the spatial coverage sporadic. The northern-most station (C) was not occupied during Deployment 7, but the instruments at the five southern moorings all worked and returned usable information.

Current velocities were separated into cross-shore (u : +, eastward) and longshore (v : +, northward) components after using a principle components analysis for each instrument which rotated the axes to orthogonal directions that minimized covariances between u and v .

Data were further transformed into a standardized form by subtracting the record mean and dividing by the standard deviation (SD). Empirical orthogonal functions (EOFs) were calculated using the standardized data. Further information on the data analysis techniques is given in Winant and Bratkovich (1981).

Temperature profile measurements made during Deployment 7 were used to calculate a maximum buoyancy frequency, N , of $2.8 \times 10^{-2} \text{ s}^{-1}$ (Jackson, 1984).

3. RESULTS

3.1. Cross-shore structure: Deployment 7

Longshore velocities decrease moving into the kelp bed from offshore (Fig. 2). The mean longshore current that was $\sim 2 \text{ cm s}^{-1}$ at O was negligible inside the kelp (Fig. 3A); the SD of 9.9 cm s^{-1} at O was larger than the 2 cm s^{-1} at K and L (Fig. 3). Within the kelp, velocities do not change in the cross-shore directions, as the mean and SD of longshore currents within the kelp bed at L and K are essentially the same. Power spectra of the longshore currents confirm the spatial patterns (Fig. 4). Power spectra are two orders of magnitude smaller at K and L than at O for sub-tidal frequencies and one order of

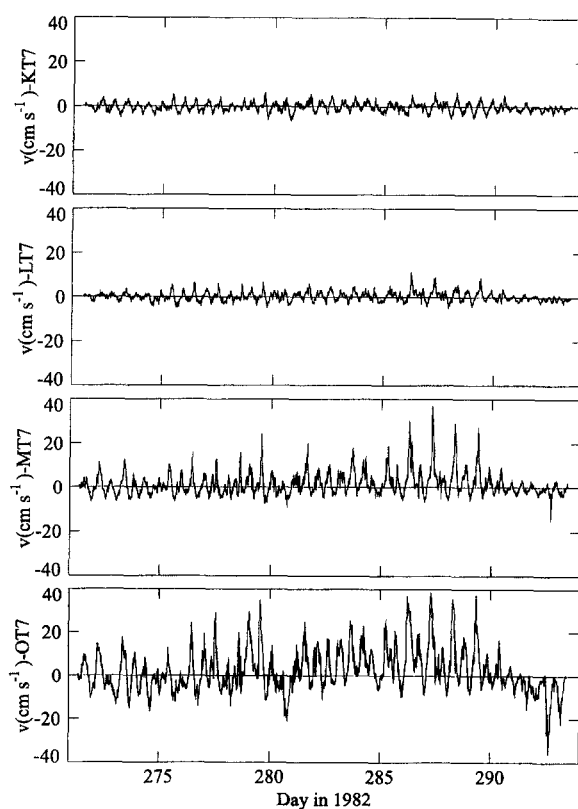


Fig. 2. Near-surface longshore velocities at K, L, M, and O during deployment 7. Values plotted represent 8' averages.

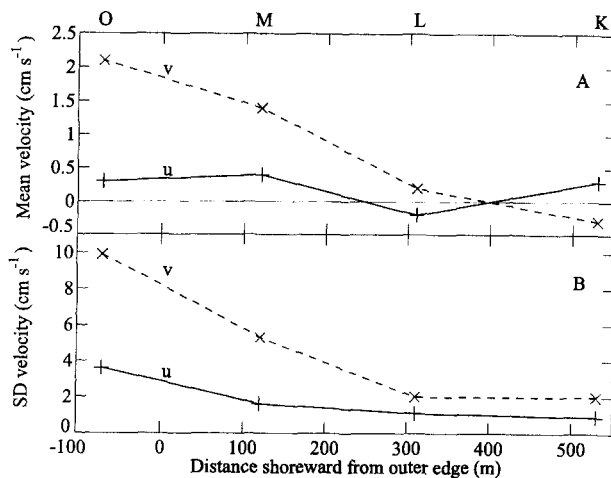


Fig. 3. Statistical properties of near-surface velocities during deployment 7 as a function of distance shoreward of outer kelp bed edge. (A) Means of *u* (cross-shore) and *v* (longshore) velocities; (B) Standard deviations of *u* and *v*.

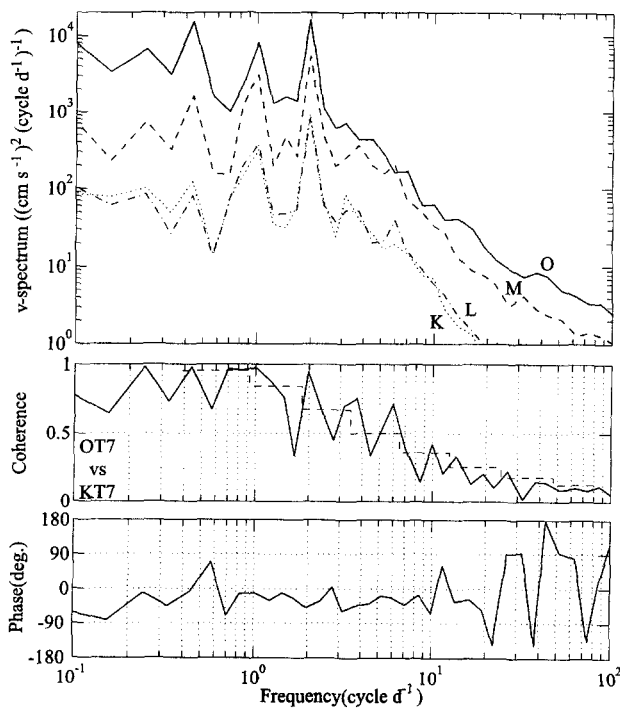


Fig. 4. Power spectra for longshore velocities at K, L, M, and O during deployment 7. Also shown are the coherence and phase between records from near-surface O and K sites. Positive phase indicates that KT lags OT.

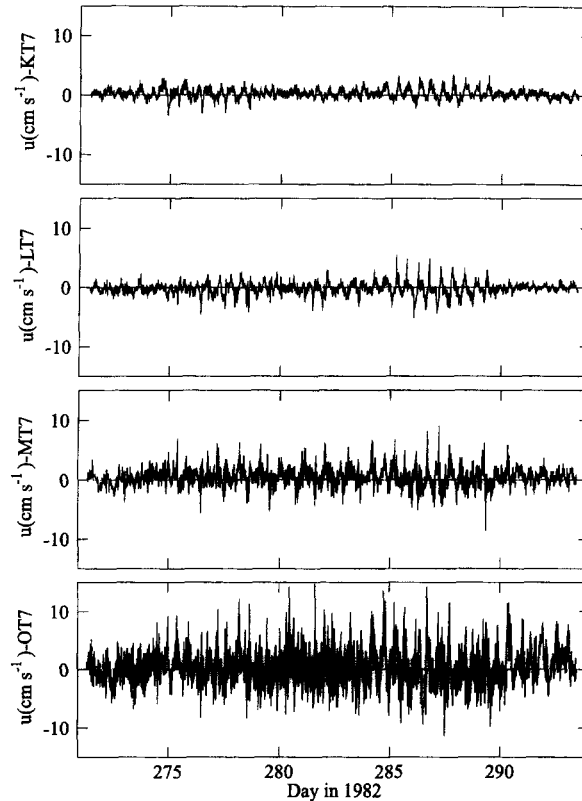


Fig. 5. Near-surface cross-shore velocities at K, L, M, and O during deployment 7. Values represent 8' averages.

magnitude smaller for supratidal frequencies. Longshore currents at all moorings have strong diurnal and semidiurnal components, with coherences of almost 1 at frequencies of 2 cycles day^{-1} . The semidiurnal components closer to shore lead those offshore, with the phase at K, L, and M relative to O being 50° , 39° , and 10° ahead of that at O. Power spectra for L and K are essentially the same.

Mean cross-shore velocities are essentially 0 (Fig. 5). The SD of cross-shore velocities decrease approximately linearly with distance into the kelp bed (Fig. 3). Power spectra of cross-shore velocities show a similar decrease in variance with decreasing distance to shore, particularly at supratidal frequencies (Fig. 6). Spectra of cross-shore velocities show a strong semidiurnal peak, with a weaker to non-existent diurnal peak.

3.1.1. Empirical orthogonal functions. The dominant EOFs for cross- and longshore velocities and temperature are essentially the average of the standardized properties (Fig. 7). They explain 0.44, 0.78, and 0.78 of the variance in u , v , and T . Cross-spectra for the first EOFs for temperature and longshore velocity are weakly coherent at diurnal and semidiurnal frequencies (not shown). They display a phase shift that is a function of frequency, with the T EOF lagging the v EOF by 180° at about 1 cycle day^{-1} , but dropping to 0° at about 5 cycles day^{-1} . The second temperature EOF and the first cross-shore current

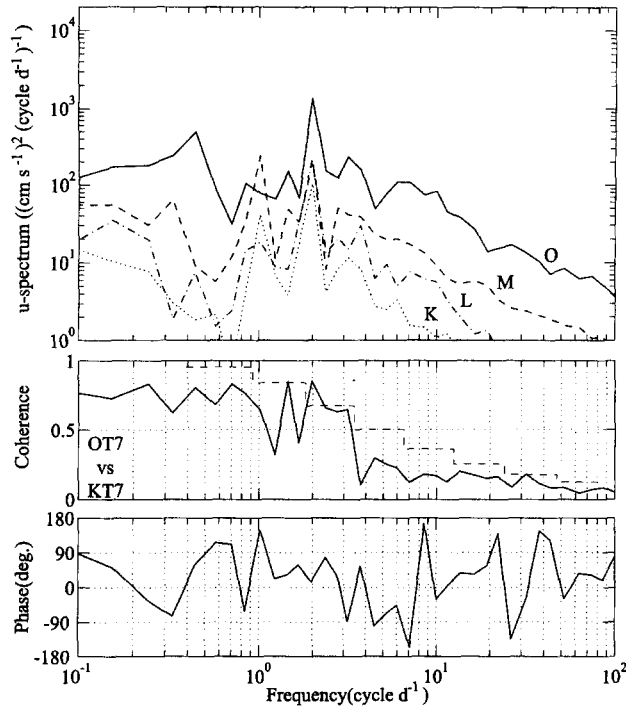


Fig. 6. Power spectra for cross-shore velocities at K, L, M, and O during deployment 7. Also shown are the coherence and phase between records from near-surface O and K sites. Positive phase indicates that KT lags OT.

EOF are related, with coherence and phase shift of about 0.8 and 0° at the dominant semidiurnal frequency.

3.2. Longshore structure: Deployment 4

Deployment 4 was designed to observe longshore relationships among physical properties, but was only partially successful because of instrumental failures. It did provide near surface records from O and L as well as for C, located 430 m inside the outer kelp bed edge and 1.3 km north of L. Mooring S was located too close to the outer edge of the kelp bed to be directly comparable to K and C. Results from S are, therefore, not shown here to avoid obscuring the presentation.

Near-surface temperature fluctuations were similar throughout the kelp bed (not shown). Temperature power spectra are similar at semidiurnal frequencies and lower, diverging at higher frequencies (Fig. 8). Moorings further into the kelp have less variance at the high frequencies. This is consistent with their control by shoreward propagation of internal waves originating offshore.

Longshore velocities at the northern C were substantially slower than at L (Fig. 9). Power spectra show similar strong activity at diurnal and semidiurnal frequencies with high coherences at both (Fig. 10). The difference is that spectral values at the northern C are about 0.1 those at the kelp bed L station, which are, in turn, about 0.1 of those outside the kelp at O (Fig. 10).

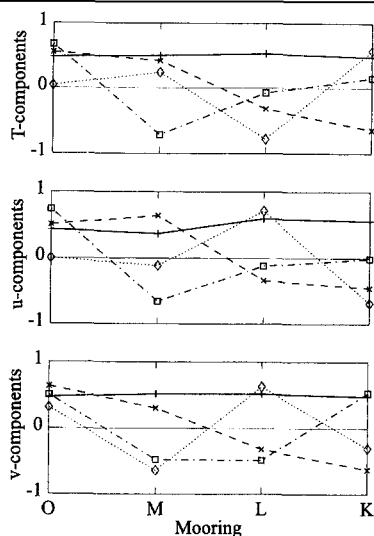


Fig. 7. The components of the EOFs calculated separately for temperature (top), cross-shore velocity (middle), and longshore velocity (bottom). The temperature EOFs account for 0.78 (—), 0.15 (—), 0.05 (—) and 0.02 (···) of the variability of the standardized temperature records; the cross-shore velocities EOFs account for 0.44 (—), 0.27 (—), 0.18 (—), and 0.12 (···) of the variability of the standardized cross-shore current records; the longshore velocity EOFs account for 0.78 (—), 0.14 (—), 0.05 (—) and 0.04 (···) of the variability of the standardized longshore velocity records.

Cross-shore velocity is related to distance from shore. Power spectra again show the dominance of diurnal and semidiurnal frequencies and decrease of energy with increasing distance into the kelp (Fig. 11), although the differences between L and C are not as great as those for longshore currents.

4. DISCUSSION

Previously published work has established that kelp beds do affect the local currents by slowing the motions (Jackson, 1983; Jackson and Winant, 1983). The present report details the spatial structure of these changes. Temporal and spatial variations in u , v , and T are different in ways that are consistent with observations in non-kelp regions (e.g. Winant and Bratkovich, 1981). Cross-spectral analysis of cross- and longshore velocities (not shown) yields surprisingly little evidence of coupling between the two. There are hints of interactions for T - u and T - v in the analyses, particularly when comparing the EOF decompositions of currents and temperature (not shown). Such relationships are expected for currents driven by density gradients.

Currents on the Southern California shelf have strong components at diurnal and semidiurnal frequencies (e.g. Winant and Olson, 1976; Winant and Bratkovich, 1981; Winant, 1983). In a detailed study of cross-shelf structure over 1 year, Winant and Bratkovich (1981) examined velocity and temperature records for a transect off Del Mar, to the north of Pt. Loma. There were current meters in 60, 30, and 15 m of water in a region with a fairly constant bottom slope. Winant and Bratkovich divided their records into three

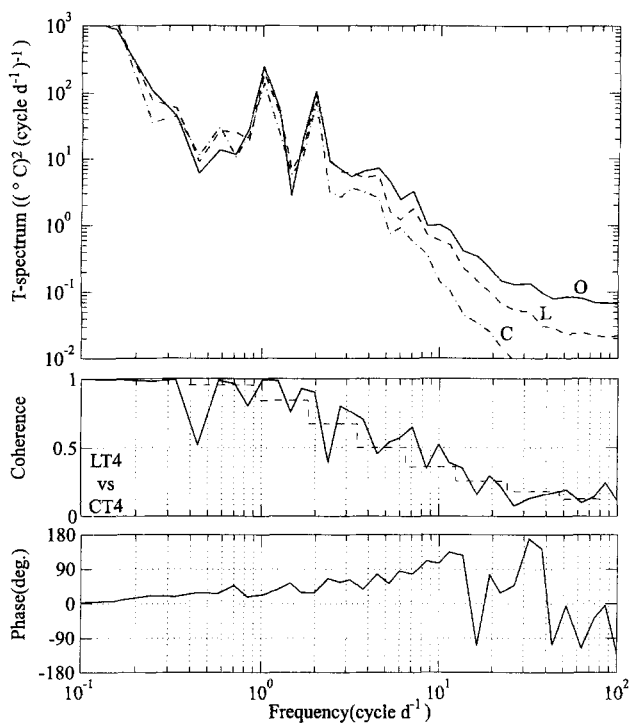


Fig. 8. Temperature spectra for near-surface records at O, L, and C. Also shown are the coherence and phase between records from near-surface L and C sites. Positive phase indicates that CT lags LT.

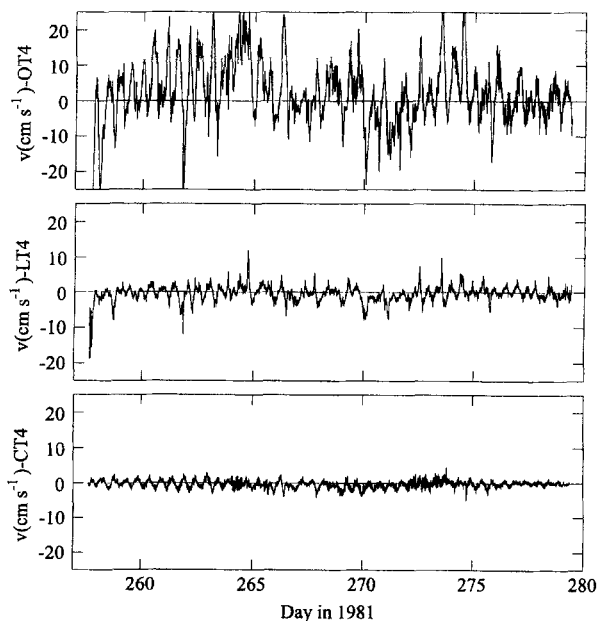


Fig. 9. Longshore current records during deployment 4. Shown are the near-surface records at O, L, and C. Values represent 8' averages.

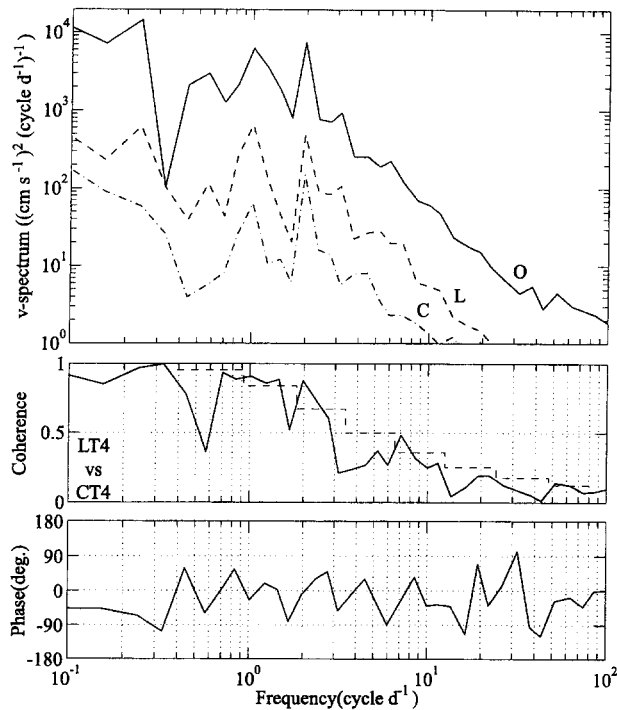


Fig. 10. Longshore velocity power spectra for near-surface records at O, L, and C. Also shown are the coherence and phase between records from near-surface L and C sites.

frequency bands: subtidal ($0.02\text{--}0.6\text{ cycles day}^{-1}$), tidal ($0.6\text{--}6\text{ cycles day}^{-1}$), and supratidal ($>6\text{ cycles day}^{-1}$). The short length of the records in the kelp bed make it difficult to compare the spatial structure of subtidal components to that at Del Mar, but the two data sets can be compared in the tidal and supratidal bands, particularly by matching data from the 30- and 15-m Del Mar moorings with those from the O and K kelp bed moorings.

4.1. Temperature

Temperature fluctuations are dominated by internal waves generated offshore and propagating shoreward in the kelp as off Del Mar (Winant and Bratkovich, 1981). There are strong signals at diurnal and semidiurnal frequencies that are not phase-locked to the surface tides (Winant and Bratkovich, 1981). Previously analyzed temperature records (Jackson, 1984) show that variations at diurnal and semidiurnal frequencies dominate within the kelp as well. The similarity in T variations at the different kelp bed moorings manifests itself with spectra that are coherent across the bed and phase differences that increase systematically with distance between moorings. Smoother records from locations further into the kelp are manifestations of increasing damping of high frequency components by the kelp. The T results have been interpreted in terms of linear internal wave theory, with waves propagating shoreward through a high-drag region of the kelp (Jackson, 1984). The phase velocities and attenuation coefficients calculated from the

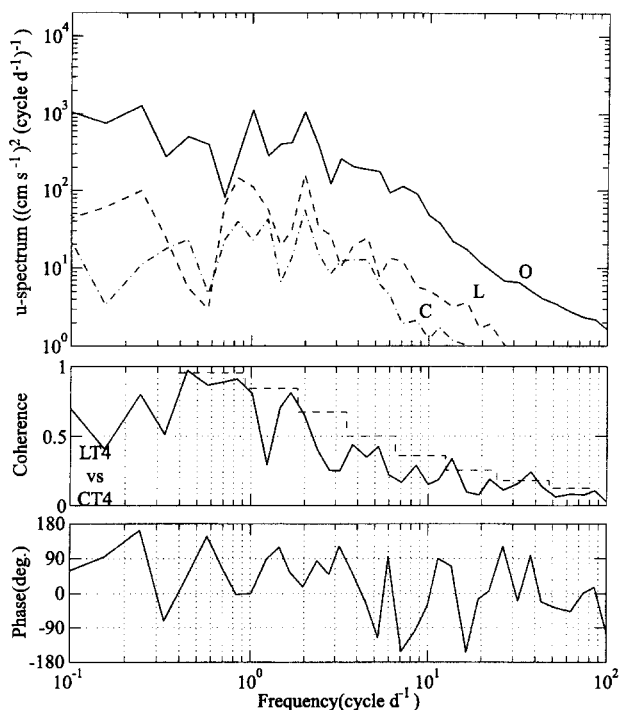


Fig. 11. Cross-shore velocity power spectra for near-surface records at O, L, and C. Also shown are the coherence and phase between records from near-surface L and C sites.

observed spectra are consistent with results calculated from the model. The differences observed at the northern C relative to that along the K–L–M–O line are consistent with the distance of C from the outer edge of the kelp bed.

4.2. Longshore velocity

Longshore velocities in shallow waters on the Southern California shelf are dominated by tidal frequency components that have a mixture of barotropic and baroclinic vertical structure and that are not correlated to changes in tidal elevations (Winant and Bratkovich, 1981). Longshore velocities along the 20-m bottom contour in the tidal band are significantly correlated for distances of about 10 km along the coast (Winant, 1983). The variances or, equivalently, the SDs of longshore velocities in the tidal band are essentially the same at 30 and 15 m and the same in winter and summer (Bratkovich, 1981). Variance in the subtidal band at 15 m is about 0.30 of that at 30 m for both summer and winter, equivalent to a SD ratio of 0.55 (Bratkovich, 1981). The decreases of longshore velocities in the kelp described here are much larger than would expected for the equivalent change in bottom depth from 30 to 15 m at Del Mar and are a clear indication of kelp modification.

Mean longshore currents within the kelp are small enough as to be unimportant for movement of material into or out of the bed. More important to such transport are the variations, such as those which would be driven by coastally-trapped waves. Jackson (1988) studied the propagation of an internal Kelvin wave along a flat bottom next to a coast with a

finite width high-drag region next to the coastal boundary. The usual propagation of a Kelvin wave along the coast with boundary to the right was modified by a slowing of the wave speed as well as a loss of energy from all areas to the high drag region. A wave propagating in the y direction decreases, with its amplitude proportional to $e^{-k_i y}$. For a semidiurnal frequency ($\omega = 1.41 \times 10^{-4} \text{ s}^{-1}$) and a drag typical of that at Pt. Loma, the theory predicts $k_i = 0.53 \text{ km}^{-1}$. A comparable value of k_i can be estimated from the data using the ratio of v SD for L and C (0.5) and the longshore separation distance of 0.8 km as $k_i = -(0.8 \text{ km}) \ln 0.5 = 0.87 \text{ km}^{-1}$.

The observed longshore velocities show several similarities to those of the trapped wave models besides the similarity in attenuation coefficients noted above: observed v spectra decrease inside the kelp bed relative to those outside, as do those in the models; observed v spectra within the bed are little different in the cross-shore direction but decrease in the longshore direction, as do those of the models; observed cross-shore velocities show little coherence with longshore velocities, as expected from the model. One prediction of the models is that the high drag of the kelp bed should dampen longshore velocities offshore as well. T. Hendricks (pers. comm.) has observed such an effect, finding a substantial decrease in v at moorings along the 35-m bottom contour off the Point Loma kelp bed with increasing distances to the north.

The ocean bottom off Pt. Loma is not flat and the distribution of the kelp is not uniform, but the correspondence between the predicted and observed changes in longshore velocities suggests that coastally trapped waves propagating to the north are important in driving them. While there can be Lagrangian motions associated with such waves, they should be smaller than those resulting from cross-shore motions.

4.3. Cross-shore velocity and movement

Winant and Bratkovich (1981) concluded that cross-shore velocity fluctuations are dominated by propagation of internal waves moving shoreward. Supporting evidence includes the correlation of T with u as well as the phase difference between u at the top and bottom of the water column. The kelp bed records show similar patterns. Furthermore, there is a systematic decrease in u with decreasing bottom depth at Del Mar (Bratkovich, 1981) that is similar that observed for the Pt. Loma kelp bed. Because of the decrease in density stratification during the winter, the variance in u at Del Mar during the winter is almost twice that during the summer in the tidal band at the three moorings. The limited data sets available in the kelp bed, however, do not show such a dramatic seasonal change in SD of u (Table 1).

The cross-shore current velocities in the kelp have SDs that are approximately linearly related to distance from shore. The dominant EOF has equal contributions from the normalized cross-shore currents along a cross-shore line, a pattern that is consistent with cross-shore motions driven by uniform vertical movements of isopycnals. There is also a hint of a trapped oscillation in the second EOF, in which the motions at the inner (K, L) moorings are out of phase with those of the outer (O, M) moorings.

The rate at which water moves from the exterior to the kelp bed is important for the functioning of the kelp ecosystem. The diminished longshore currents and the greater length of the kelp bed alongshore make them relatively unimportant in providing the exchange (e.g. Jackson, 1983, 1988). However, the small correlations for u between moorings makes it difficult to interpolate particle trajectories between them. The EOFs provide a means to

examine cross-shore exchange. The approximately linear dependence with distance offshore of the SDs of u , which are used to normalize the values of u , and the equal components of the first EOF imply that the cross-shore velocity associated with it, u_1 , is linearly related to the distance offshore x

$$u_1 = a(t)x \quad (1)$$

If $e_{u,1}(t)$ is the value of the first EOF as a function time, then

$$u_1(t) = \frac{\sigma}{L} e_{u,1}(t)x \quad (2)$$

where σ is the standard deviation at $x = L$. Integrating $u = \frac{dx}{dt}$ yields

$$x = x_0 e^{\frac{\sigma}{L} \int_0^t e_{u,1}(t) dt} \quad (3)$$

Equation (3) can be used to estimate the excursions of water into the bed. The maximum movement water initially at the outer edge of the bed would have made in 24 h has been calculated and this showed that excursions greater than 400 m are frequent (Fig. 12). Excursions 600 m or greater are episodic at best.

4.4. Ecological significance

The damping of water motions from the surrounding ocean is reminiscent of boundary layer effects in terrestrial ecosystems, the study of which goes under the name micrometeorology. In forests, the high drag of the trees causes significant decreases in wind velocity and turbulent energy dissipation below the canopies as trees substantially alter

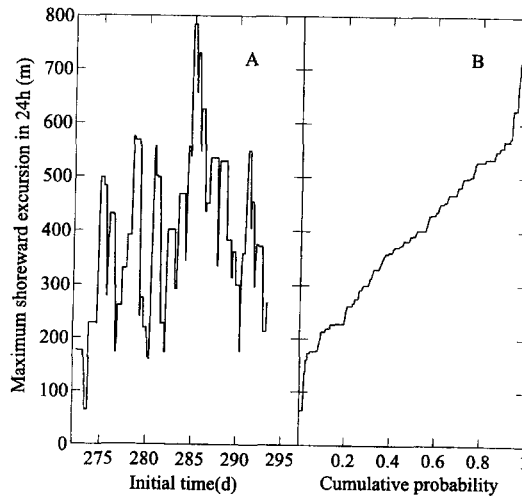


Fig. 12. (A) Distance into the kelp bed water on the outer edge of the kelp bed would go in 24 h during deployment 7; (B) Cumulative probability of excursions less than a given distance. The calculation assumes that the water goes no further offshore than the outer edge of the bed at $L = -1000$ m and that $\sigma = 0.02 \text{ m s}^{-1}$. Excursions of 0 m in A indicate that u_1 is offshore or 0 for the 24-h period.

the planetary boundary layer and shelter each other from the force of the wind. There are similarities and differences to the microoceanography of a kelp bed.

Trees on the outer edge of a forest are exposed to the brunt of strong winds and are more vulnerable to being blown down (e.g. Peltola, 1996). This is particularly a problem for trees bordering a clear-cut region. The equivalent effect of a strong current on kelp plants is to submerge them, removing photosynthetic tissue from the high light environment near the surface. The damping of longshore currents as coastal waves move upcoast (north on the west coast of North America) implies that plants in the interior of a kelp bed are less subject to such submergence than those at the leading edge and that the alongshore length of a bed can be an important property in determining plant submergence.

Another important effect of wind motion in a forest or in agricultural land is to enhance the transport of water and carbon dioxide to and from plants by decreasing boundary layers around individual leaves. Water motion has a similar effect on nitrate and carbon dioxide transport and uptake by blades in the related kelp *Macrocystis integrifolia*, for which nutrient uptake rates are transport-limited at water velocities less than 6 cm s^{-1} (Wheeler, 1980; Hurd *et al.*, 1996). Much of the Pt. Loma kelp bed is in water with velocities less than 5 cm s^{-1} , suggesting that the decreased water velocities associated with a large kelp bed could slow plant growth rates.

Surface gravity waves can also provide water motion capable of overcoming transport limitation. While the effects of these surface waves vary with depth and wave energy spectrum, the interactions of a plant with wave-induced water motion are quite different than with a longshore current. Seymour (1996) has noted that the flexibility and length of giant kelp allow plants to move with the waves, resulting in little relative water motion. However, Gerard (1982) has argued that normal wave motions are sufficient to overcome any boundary layer transport limitation. It is, thus, unclear the extent to which wave motions can overcome any transport limitation for vital nutrients.

An effect which has wide implications for the associated ecosystem is the decrease in transport of material into or out of a kelp bed. For a forest, the concerns are for transport of water vapor and carbon dioxide (Monteith and Unsworth, 1990). Aside from the region near the edge of a forest, material exchange with the atmosphere is predominantly vertical. In a kelp bed, the role of the vertical dimension is replaced with the cross-shore.

Exchange from longshore currents appears to be greatest in a region of a few 100 m at the upstream end of the kelp bed (Jackson, 1984). Further longshore transport decreases as net currents are diverted around the bed and longshore motions are damped by the high drag of the kelp. There are frequent incursions of water from the offshore edge for distances of about 500 m from the outer edge, but fewer for greater distances. For a kelp bed with a width of 1 km, such as that at Pt. Loma, movement across the bed to the inshore edge are less frequent.

Cross-shore flow should be sensitive to density stratification that would allow internal waves to propagate. Hence, there should be less cross-shore movement during the winter. This analysis ignores other mechanisms of cross-shore transport, such as that driven by upwelling.

While the discussion here and in Jackson (1984) have addressed the issue of exchange at the upstream end and outer edges of the kelp, they have not provided information about the movement of water from the inshore edge. The inshore edge of the kelp is in about 6–8 m of water in a region where wave forces can tear out plants (Seymour *et al.*, 1989) and where waves affect longshore movement of water (e.g. Inman *et al.*, 1971). This cleared region

provides an alternate channel for movement of water to the kelp bed. The loss of stratification at shallowest depths should inhibit movement of material into a kelp bed by this mechanism. For kelp beds which have an inner edge in shallow water, barring another transport mechanism, expect that inshore edge of kelp will be less important for moving material inside.

The movement of external water bearing food (e.g. Bray, 1981; Anderson and Sabado, 1995), larvae (e.g. Bernstein and Jung, 1979; Gaines and Roughgarden, 1987; Schroeter *et al.*, 1996), and nutrients (e.g. Jackson, 1977) has long been understood to be important to the functioning of kelp ecosystems. The resulting distributions of food, larvae, and nutrients depend on the dynamics of both their physical transport and the biological processes decreasing them (e.g. Jackson, 1986). For fish feeding on zooplankton, the biological predation rates can be quite large (e.g. Bray, 1981; Gaines and Roughgarden, 1987; Anderson and Sabado, 1995). Evidence from larval settlement is more ambiguous.

Schroeter *et al.* (1996) looked for variations in sea urchin recruitment as a function of position in the Pt. Loma kelp bed. Sea urchin larvae are planktonic and should enter the kelp bed from offshore waters. Schroeter *et al.* found differences in larval recruitment between the inside and outside edges of the kelp which were suggestive but not statistically significant. Interpretation of their results is confounded by the lack of a station in the kelp bed interior. Larval transport through the inshore wave zone cannot be ruled out as a source of larvae on the inshore edge of their stations. Furthermore, spatial patterns of larval settling depend on biological reactions to suitable habitat as well as on mass transport rates (e.g. Jackson, 1986). While transport may not be the dominant factor for the sea urchin species studied, it could be for species with different behavioral responses.

5. CONCLUSION

Kelp beds have unique physical environments as a result of the high drag. Damping of the longshore currents that are usually most energetic in shallow coastal regions minimizes the effect that they can have on transport of material into the bed and that they can exert to overcome diffusional transport limitation to blades. Cross-shore currents can provide significant movement of material to the interior. The ecological importance of decreased water movement depends on biological factors as well as physical.

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